CS683 Project Proposal

Runtime Context based Set Dueling for Cache Replacement

Mohammed Nazal
26M0752@iitb.ac.in

Problem statement

Evaluating whether lightweight runtime context improves Set Dueling's policy selection while keeping the hardware mechanism simple.

Prior Works

- Retrofitting Contextual Learning into Set Dueling (*D. Patsidis, L. Peled, G. Vavouliotis*, MICRO 26)
- Context-Aware Set Dueling (CASD) *(G. Vavouliotis et al CAL 2025)*

Goal of the Project

- Study whether runtime workload context such as MPKI can improve cache replacement policy selection with lesser hardware overhead.
- Simulator - **ChampSim**

- Implementations, experiments and final project work will be my own effort. Gen-AI will not be used for project implementation or experiments.

Plan for checkpoint-I

1) Experiments :
   - Validate replacement policy and Set Dueling in Champsim and Evaluate them on Champsim traces.
   - Analyse runtime LLC MPKI and workload behaviour

1) Insights expected to find :
   - Difference in behaviours of replacement policies
   - If policy preference changes during execution of certain workloads

Plan for checkpoint-II

1. Experiments
    a. Implement runtime context based Set Dueling using LLC MPKI
    b. Evaluate and compare it with conventional Set Dueling on ChampSim traces.
2. Insights expected to find
    a. IF runtime context improves policy selection in Set Dueling